Part 4 Brief · Evals

mcp-ops-server · 18 Oct 2026 · PR #5 · Status: Codex review fixed (5x P1, 5x P2); 31/31 tasks, 12/12 controls; ready to merge

WHAT WAS BUILT

- **31 golden tasks** in 7 categories (lookups, bookings, referral bookings, refusals, cancellations, auth, conflicts). Each has a prompt, who it runs as (staff, read-only, or a patient-bound portal token) and what must and must not happen.
- **A trajectory grader** that checks every step, not just the final answer: required steps in order, forbidden steps, no touching other patients, call and error budgets, and the **exact change in the database**.
- **A replay agent** (no LLM, free, deterministic) that runs in CI, plus `evals/report.md`: per-task results, pass rate by category, tool latency.
- **12 negative controls:** deliberately bad runs, including adversarial ones that try to game the grader (checking the wrong referral, booking then cancelling to hide it). Each must fail.

HOW IT WORKS

Piece	Role
harness.py	A fresh seeded database and real server per task (same app as the CLI, real auth), clock pinned. Records every call the agent makes and diffs the appointments table.
grader.py	Eight checks per task, on the recorded calls (not the agent's report) and on the exact database change, so a correct-sounding but wrong run still fails.
agents.py	One Agent interface. Replay now; the Part 5 LLM agent plugs into the same slot and is graded identically.

WHY IT WAS DONE THIS WAY

- **Grade the database, not the reply.** An agent can say "booked!" and book the wrong patient. The diff catches that; a final-answer check would not.
- **Grade the steps.** Booking with a valid referral ID but never checking its status can still succeed. Only step-level grading sees that the check was skipped.
- **Evals use real auth.** 8 of the 66 calls are expected refusals (scope, patient binding, double booking, past slot), so the suite also tests Part 3's rules end to end.

WHAT WENT WRONG OR SURPRISED ME (LOGGED IN FAILURES.MD)

- **A loosened rule was invisible to every golden task.** I made pending referrals bookable as a test; all 31 tasks still passed, because a well-behaved agent never tries it. Only a negative control failed. Now the controls must cover every check.
- **Codex review: the grader could be fooled 9 ways.** Agents reported their own steps; "earliest" was checked by date only; any referral check counted; book-then-cancel vanished; one bad run poisoned the next. All fixed, each with a test or control that fails without the fix.

VERIFY IT YOURSELF

```
python -m evals.run       # 31/31 tasks passed; 12/12 negative controls caught
cat evals/report.md       # per-task results, categories, latency
ruff check . && python -m pytest -q   # 159 passed (includes the full suite)
```

INTERVIEW TALKING POINT

"My eval grades what the agent actually did, recorded by the harness, and what changed in the database, not what it said. Then I attacked my own grader: an independent review found 9 ways to pass a bad run, and each is now a negative control that must fail."

NEXT: PART 5, CI GATE AND REAL NUMBERS

A dedicated eval job that blocks merges, and a live LLM agent on the same 31 tasks to measure pass rate, cost and latency. That needs an API key as a secret, your call. **Your action:** merge PR #5, then reply "go" with the model, API and run frequency for the live agent.